Jian Weng

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Education

University of California, Los Angeles **Los Angeles, CA**
Ph.D. in Computer Science, Advisor: Tony Nowatzki *Aug. 2017 – Sep. 2023*
Thesis: Developing, Synthesizing, and Automating Domain-Specific Accelerators

Shanghai Jiao Tong University **Shanghai, China**
B.Eng. in Computer Science, ACM Honored Class *May 2013 – July 2017*
Thesis: Compiling Domain Specific Language to Reconfigurable Accelerators

Professional Experiences

King Abdullah University of Science and Technology **Thuwal, Mecca Province**
Assistant Professor *Jan. 2024 — Present*

Conference Publications

- AQuant: Repurposing CODEC for VLM Acceleration via Adaptive Quantization.
  Zhuoran Song, Chunyu Qi, **Jian Weng**, Xiaoyao Liang, Haibing Guan.
  International Symposium on Computer Architecture (ISCA), 2026
- Hierarchical Exploitation of Memory Efficiency for GPU-Based FHE Acceleration.
  Guang Fan, Yi Chen, Lei Chen, Liang Kong, Chao Niu, Dian Jiao, Yilan Zhu, Geng Yang, Shengyu Fan, Xianglong Deng, Fangyu Zheng, **Jian Weng**, Yisong Chang, Meng Li, Shoumeng Yan, Mingzhe Zhang.
  International Symposium on Computer Architecture (ISCA), 2026
- MLX: Multi-Layer Execution for Structured LLM Workload Acceleration on Spatial Architectures.
  Haibin Wu, Wenming Li, Zhihua Fan, Zirui Ma, Yuqun Liu, Tengfei Xia, Yanhuan Liu, Kunming Zhang, Xiaochun Ye, Dongrui Fan, **Jian Weng**.
  International Symposium on Computer Architecture (ISCA), 2026, **Best Paper Candidate**
- SpecVLA: Enabling Efficient Robotic Manipulation via Speculative VLA Inference and Verification.
  Chunyu Qi, Zhuoran Song, **Jian Weng**, Haozhe Jiang, Xueyuan Liu, Naifeng Jing, Guanghui He, Xiaoyao Liang, Haibing Guan.
  International Symposium on Microarchitecture (MICRO), 2026
- GPU-Tile-Sim: A Tile-Centric GPU Simulation Framework for LLM Hardware-Software Co-Design.
  Yitong Ding, Jiawei Huang, Renyang Guan, Zihan Liu, Yangjie Zhou, Yu Feng, Shixuan Sun, Jingwen Leng, Mingyi Guo, **Jian Weng**.
  International Symposium on Microarchitecture (MICRO), 2026
- Assassyn: A Unified Framework for Architecture Synthesis and Simulation.
  Jian Weng, Boyang Han, Derui Gao, Ruijie Gao, Wanning Zhang, An Zhong, Ceyu Xu, Jihao Xin, Yangzhixin Luo, Lisa Wu Wills, Marco Canini.
  International Symposium on Computer Architecture (ISCA), 2025
- OverGen: Improving FPGA Usability through Domain-specific Overlay Generation.
  Sihao Liu=, **Jian Weng=**, Dylan Kupsh, Atefeh Sohrabizadeh, Zhengrong Wang, Licheng Guo, Jiuyang Liu, Maxim Zhulin, Lucheng Zhang, Jason Cong, Tony Nowatzki.
  International Symposium on Microarchitecture (MICRO), 2022, **Best Paper Runner-up**
- Near-Stream Computing: General and Transparent Near-Cache Acceleration.
  Zhengrong Wang, **Jian Weng**, Sihao Liu, Tony Nowatzki.
  International Symposium on High-Performance Computer Architecture (HPCA), 2022

- UNIT: Unifying Tensorized Instruction Compilation.
  Jian Weng, Animesh Jain, Jie Wang, Leyuan Wang, Yida Wang, and Tony Nowatzki.
  International Symposium on Code Generation and Optimization (CGO), 2021
- Stream Floating: Enabling Proactive and Decentralized Cache Optimizations.
  Zhengrong Wang, **Jian Weng**, Jason Lowe-Power, Jayesh Gaur, Tony Nowatzki. *International Symposium on High-Performance Computer Architecture (HPCA), 2021*, **Best Paper Runner-up**
- DSAGEN: Synthesizing Programmable Spatial Accelerators.
  Jian Weng=, Sihao Liu=, Vidushi Dadu , Zhengrong Wang , Preyas Shah, and Tony Nowatzki.
  International Symposium on Computer Architecture (ISCA), 2020, **IEEE Micro Honorable Mentions**
- A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms.
  Jian Weng, Sihao Liu, Zhengrong Wang, Vidushi Dadu, and Tony Nowatzki.
  International Symposium on High-Performance Computer Architecture (HPCA), 2020
- Towards General Purpose Acceleration by Exploiting Common Data-Dependence Forms.
  Vidushi Dadu, **Jian Weng**, Sihao Liu, and Tony Nowatzki.
  International Symposium on Microarchitecture (MICRO), 2019 **IEEE Micro Top Picks**
- Hybrid Optimization/Heuristic Instruction Scheduling for Programmable Accelerator Codesign.
  Tony Nowatzki, Newsha Ardalani, Karthikeyan Sankaralingam, and **Jian Weng**.
  Parallel Architectures and Compilation Techniques (PACT), 2018

Journal Publications

- BISC: Rethinking Instruction Set Abstractions at Block Granularity.
  An Zhong, Hanzhi Hu, Zirui Ma, Mengzhu Li, Haibin Wu, Fulin Zhang, Jinyi Deng, Ceyu Xu, Wenming Li, Yang Hu, **Jian Weng**
  IEEE Computer Architecture Letters, 2026
- CACO: Chiplet-based FHE Acceleration with Inter-Chiplet Communication Optimizations.
  Xianglong Deng, Shengyu Fan, Liang Kong, Yilan Zhu, Geng Yang, Lei Chen, Guang Fan, Yi Chen, Yisong Chang, **Jian Weng**, Shoumeng Yan, Mingzhe Zhang
  IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026
- PIMSAB: A Processing-In-Memory System with Spatially-Aware Communication and Bit-Serial-Aware Computation.
  Siyuan Ma, Kaustbh Mhatre, **Jian Weng**, Bagus Hanindhito, Zhengrong Wang, Tony Nowatzki, Lizy John, Aman Arora
  ACM Transactions on Architecture and Code Optimization, 2024
- Unifying Spatial Accelerator Compilation with Idiomatic and Modular Transformations.
  Jian Weng, Sihao Liu, Dylan Kupsh, Tony Nowatzki.
  IEEE Micro Special Issue on Compiling for Accelerators, 2022
- DAEGEN: A Modular Compiler for Exploring Decoupled Access Execte Accelerators.
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  Computer Architecture Letters (CAL), 2019

Workshops and Tutorials

- OverGen: Decoupled-Spatial Architecture Framework
  Sihao Liu, **Jian Weng**, Dylan Kupsh, and Tony Nowatzki.
  International Symposium on Microarchitecture (@MICRO'22)
- A Full-Stack Infrastructure for Automating Spatial Architecture Research
  Jian Weng, Sihao Liu, Dylan Kupsh, and Tony Nowatzki.
  Workshop on Democratizing Domain-Specific Accelerators (@MICRO'22)

- Generality is the Key Dimension in Accelerator Design
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  Workshop on Languages, Tools, and Techniques for Accelerator Design (@ASPLOS'21)
- DSAGEN: Democratizing Decoupled Spatial Architecture Research
  Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
  International Symposium on Microarchitecture (@MICRO'20)

Open-Source Projects and Infrastructures

Democratizing Spatial Accelerator Design ↗
Co-first Author *Aug. 2017 – Now*

- Develop the compiler infrastructure for spatial architectures.

TVM ↗ **Apache**
Committer *Aug. 2017 – Now*

- Implement the `tvm.te.hybrid` module to enhance the expressiveness of the framework.
- Write tutorials on performance tuning and customized compilation pass.

Selected Awards

ISCA 2026 Best Paper Candidate	*2026*
MICRO 2022 Best Paper Runner-up	*Oct. 2022*
UCLA Disseration Year Fellowship	*Sep. 2022*
QualComm Fellowship Finalist	*Aug. 2022*
Facebook Fellowship Finalist	*April 2022*
UCLA Graduate Student Fellowship	*Nov. 2017, Mar. 2021*
IEEE Micro Top Picks Honorable Mention	*2021*
HPCA Best Paper Runner-up	*2021*
IEEE Micro Top Picks	*2020*

Teaching

CS294Z: Agentic Programming for Compilation **KAUST**
Instructor *2026 Spring*

CS394A: Programming Hetergeneous Systems **KAUST**
Instructor *2025 Spring*

CS398: Graduate Seminar **KAUST**
Instructor, w/ TA: Jihao Xin *2025 Spring*

Community Services

IEEE/ACM International Symposium on Microarchitecture (MICRO) **2026**
Program Committee

International Conference on Application-specific Systems, Architectures and Processors (ASAP) **2026**
Program Committee

Advanced Parallel Programming Technologies (APPT) **2026**
Publicity Chair

ACM ASPLOS, Summer **2025**
Reviewer

ACM ASPLOS, Fall **2025**
Reviewer

IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) **2025**
Program Committee

Design Automation Conference (DAC) **2025**
Program Committee

IEEE/ACM International Symposium on Computer Architecture (ISCA) **2025**
Program Committee

IEEE International Symposium on High Performance Computer Architecture (HPCA) **2025**
Program Committee

IEEE/ACM International Symposium on Microarchitecture (MICRO) **2025**
Program Committee

ACM Transactions on Architecture and Code Optimization (TACO) **2025**
Invited Reviewer

Advanced Parallel Programming Technologies (APPT) **2025**
Submission Chair

Great Lakes Symposium on VLSI (GLVLSI) **2025**
Program Committee

International Conference on Application-specific Systems, Architectures and Processors (ASAP) **2025**
Program Committee

IEEE Transactions on VLSI **2025**
Invited Reviewer

Computer Architecture Letter **2024**
Invited Reviewer

International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) **2024**
Program Committee

Design Automation Conference (DAC) **2022**
Invited Reviewer

IEEE/ACM International Symposium on Microarchitecture (MICRO) **2022**
Student Reviewer